SCIENTIFIC MACHINE LEARNING PSET02.

Problem 1.

(a) * $\delta_j^{(L)} = \dfrac{\partial c}{\partial z_j^{(L)}}$

$$= \frac{\partial c}{\partial a_j^{(L)}} \frac{\partial a_j^{(L)}}{\partial z_j^{(L)}}$$

Note that $a_j^{(L)} = g_L(z_j^{(L)})$ so $\dfrac{\partial a_j^{(L)}}{\partial z_j^{(L)}} = g_L'(z_j^{(L)})$

$$\delta_j^{(L)} = \frac{\partial c}{\partial a_j^{(L)}} g_L'(z_j^{(L)})$$

* $\delta_j^{(l)} = \dfrac{\partial c}{\partial z_j^{(l)}}$

$$z_j^{(l)} = \sum_{i=1}^{n_{l-1}} a_i^{(l-1)} W_{ij}^{(l)} + b_j^{(l)}$$

$$\frac{\partial z_j^{(l+1)}}{\partial z_k^{(l)}} = \sum_{i=1}^{n_l} \frac{\partial g_l(z_i^{(l)})}{\partial z_k} W_{ij}^{(l+1)}$$

$$= \sum_{i=1}^{n_{l-1}} g_l'(z_i^{(l)}) W_{ij}^{(l+1)} \delta_{ik}$$

$$= g_l'(z_k^{(l)}) W_{kj}^{(l+1)}$$

$$= g_l'(z_k^{(l)}) (W_{jk}^{(l+1)})^T$$

$$\delta_j^{(l)} = \sum_{k=1}^{n_{l+1}} \frac{\partial c}{\partial z_k^{(l+1)}} \frac{\partial z_k^{(l+1)}}{\partial z_j^{(l)}}$$

$$\delta_j^{(l)} = \sum_{k=1}^{n_{l+1}} \delta_k^{(l+1)} g_l'(z_j^{(l)}) (W_{kj}^{(l+1)})^T$$

l-1 layer does not depend on bias of layer l

(b) * $$\frac{\partial c}{\partial b_j^{(l)}} = \frac{\partial c}{\partial z_j^{(l)}} \cancelto{1}{\frac{\partial z_j^{(l)}}{\partial b_j^{(l)}}} + \sum_{i=1}^{n_{l-1}} \left[\frac{\partial c}{\partial a_i^{(l-1)}} \cancelto{0}{\frac{\partial a_i^{(l-1)}}{\partial b_j^{(l)}}} + \frac{\partial c}{\partial W_{ij}^{(l)}} \cancelto{0}{\frac{\partial W_{ij}^{(l)}}{\partial b_j^{(l)}}} \right]$$

$$= \delta_j^{(l)}$$

* $$\frac{\partial c}{\partial W_{ij}^{(l)}} = \frac{\partial c}{\partial z_j^{(l)}} \frac{\partial z_j^{(l)}}{\partial W_{ij}^{(l)}} + \sum_{k=1}^{n_{l-1}} \left[\frac{\partial c}{\partial a_k^{(l-1)}} \cancelto{0}{\frac{\partial a_k^{(l-1)}}{\partial W_{ij}^{(l)}}} \right] + \frac{\partial c}{\partial b_j^{(l)}} \cancelto{0}{\frac{\partial b_j^{(l)}}{\partial W_{ij}^{(l)}}}$$

l-1 layer does not depend on weight of layer l.

$$\frac{\partial z_j^{(l)}}{\partial W_{ij}^{(l)}} = a_i^{(l-1)}$$

so $$\frac{\partial c}{\partial W_{ij}^{(l)}} = \delta_j^{(l)} a_i^{(l-1)}$$